

E.HYBRID WAN

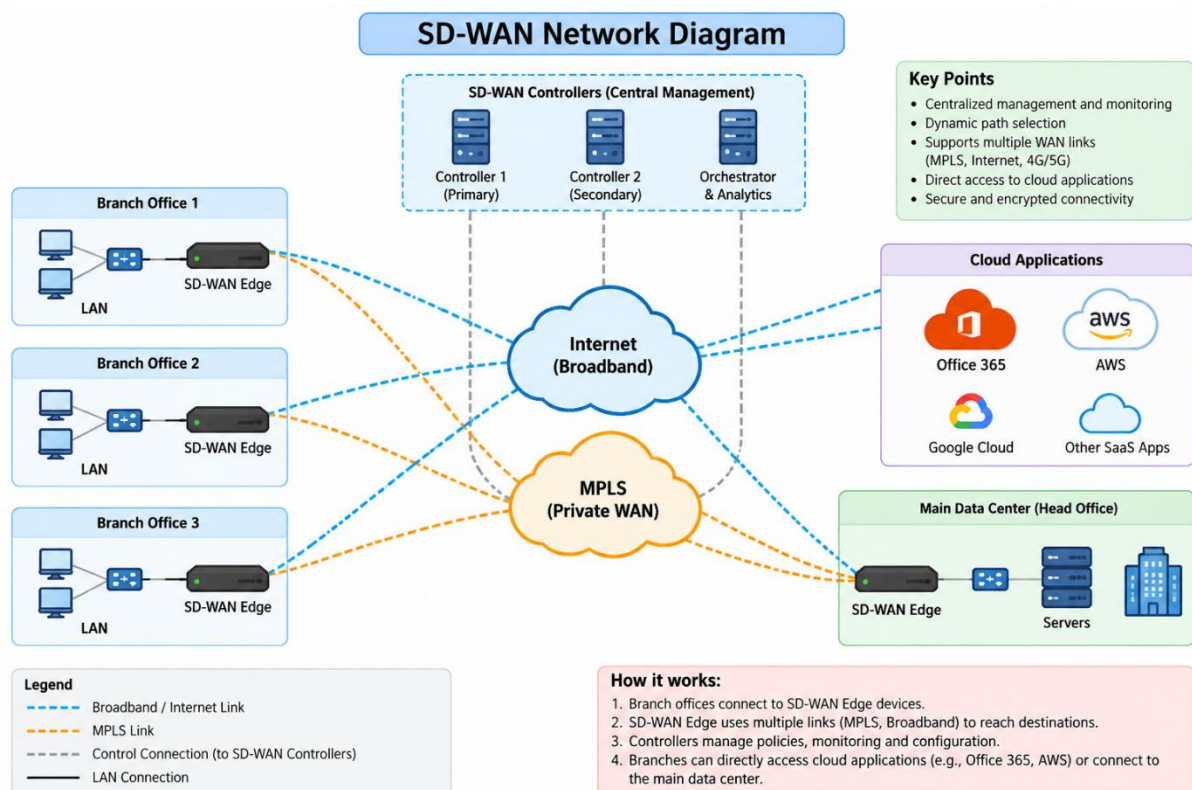
1.Create a comparison table listing at least 5 key differences between traditional WAN and SD-WAN, focusing on how data is routed, managed, and secured.

Feature	Traditional WAN	SD-WAN
1. Data Routing	Traffic usually follows fixed, pre-configured routes through routers.	Traffic is dynamically routed based on application needs, network conditions, and policies.
2. Management	Each router/device is configured and managed individually.	Centrally managed using an SD-WAN controller/orchestrator.
3. Path Selection	Usually depends on static routes or traditional routing protocols such as OSPF, EIGRP, or BGP.	Automatically selects the best available path using real-time network conditions.

4. Security	Security often depends on separate firewalls, VPNs, and security appliances at branch locations.	Security policies can be centrally configured, with features such as encryption, segmentation, and integrated security.
5. Internet Usage	Often relies heavily on expensive private WAN connections such as MPLS.	Can intelligently use multiple links such as MPLS, broadband, and 4G/5G.
6. Configuration	Changes may require manual configuration at multiple branch routers.	Policies can be configured centrally and pushed to multiple branches automatically.
7. Application Awareness	Routing is primarily based on IP addresses and routing information.	Can identify applications and prioritize critical traffic, such as VoIP or video conferencing.

8. Scalability	Adding new branches can require significant manual configuration and deployment.	New branches can be deployed more quickly using centralized configuration and automation.
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2. Draw a simple network diagram (on paper or using a tool like draw.io) showing how SD-WAN connects multiple branch offices to cloud apps (like Office 365 or AWS) and the main data center. Label the SD-WAN controllers, edges, and at least two types of WAN links (e.g., MPLS, broadband).



3. Pick any app you use daily (like WhatsApp, Zomato, or Instagram) and write a short scenario explaining how SD-WAN could improve user experience for that app when used across multiple cities in

India.
Hint: Think about speed, reliability, and security when users access the app from different locations.

Scenario: WhatsApp Using SD-WAN Across Multiple Cities in India

Imagine WhatsApp has offices and servers across different cities such as Ahmedabad, Mumbai, Delhi, Bengaluru, and Hyderabad.

Without SD-WAN, network traffic may use a fixed path. If that connection becomes slow or unstable, WhatsApp services for employees in that city may become slower.

With SD-WAN, the company can use multiple WAN connections such as MPLS, broadband, and 4G/5G.

Example: An employee in the Ahmedabad office is using WhatsApp to access company services. If the broadband connection becomes slow, SD-WAN can automatically select a better available path, such as MPLS or another Internet connection.

Benefits

- Speed: SD-WAN can select a better-performing network path.
- Reliability: If one WAN link fails, traffic can use another available link.
- Security: Traffic can be encrypted and security policies can be applied centrally.
- Better user experience: Users in different cities can get more consistent access to applications.
- Centralized management: Network administrators can manage and monitor connections across all cities from one place.

In short: SD-WAN can help an organization provide faster, more reliable, and secure access to applications across multiple Indian cities.

4. List and explain three key benefits of SD-WAN for a company with offices in Ahmedabad, Mumbai, and Bangalore. For each benefit, give a real-world example of how it would help employees or customers.

Benefit	Explanation	Real-world example
1. Better Speed & Performance	SD-WAN automatically chooses the best available network path based on network conditions and application requirements.	Employees in the Ahmedabad office are using Microsoft Teams for a video meeting with the Bangalore team. If one Internet link is slow, SD-WAN can route the traffic through a better link, reducing lag and interruptions.
2. High Reliability & Automatic Failover	SD-WAN can use multiple WAN connections such as MPLS, broadband, and 4G/5G. If one link fails, traffic can move to another available link.	The Mumbai office's broadband connection fails during an important customer transaction. SD-WAN automatically switches traffic to the backup MPLS/4G connection, allowing employees to continue working.
3. Centralized Security & Management	Network administrators can manage routing and security policies for all offices from a central SD-WAN platform.	The company can apply the same security policy to Ahmedabad, Mumbai, and Bangalore. Administrators can monitor all three offices centrally instead of configuring every router separately.

